Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/21/2005
PI Name:	Ramachandran, Narayanan Ph.D.		
Project Title:	STUDY OF FLUID FLOW CONTROL IN PROTEIN CRYSTALLIZATION USING STRONG MAGNETIC FIELDS		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	narayanan.ramachandran@msfc.nasa.gov	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	256-544-8308
Organization Name:	BAE Systems Analytical Solutions Inc.		
PI Address 1:	XD 42, NASA RAES Contract, MSFC		
PI Address 2:	308 Voyager Way		
PI Web Page:			
City:	Huntsville	**State:**	AL
Zip Code:	35806	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	05/01/2003	**End Date:**	05/01/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Niederhaus, Charles	**Contact Phone:**	(216) 433-5461
Contact Email:	charles.niederhaus@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Leslie, Fred (NASA MSFC) Ciszak, Ewa (NASA MSFC)		
Grant/Contract No.:	NCC 3 1090		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	An important component in biotechnology, particularly in the area of protein engineering and rational drug design is the knowledge of the precise three-dimensional molecular structure of proteins. The quality of structural information obtained from X-ray diffraction methods is directly dependent on the degree of perfection of the protein crystals. As a consequence, the growth of high quality macromolecular crystals for diffraction analyses has been the central focus for biochemists, biologists, and bioengineers. To date, space protein crystallization experiments have yielded inconclusive results with ~20% of flight-grown proteins showing promising results. The space crystal growth environment offers reduced fluid motion – convection due solutal buoyancy and sedimentation effects. We postulate that we can reduce convection in ground-based experiments and we can understand the different aspects of convection control through the use of strong magnetic fields and field gradients. In 2004, progress was made on several fronts in the project. A key finding based on our measurements of the density and susceptibility variation of protein solutions as a function of protein concentration showed that an equivalent g-level of only ~0.2 go (where go is terrestrial gravity) is realizable in a practical, high field strength superconducting magnet. This is because the protein solution is weakly diamagnetic (repelled by a magnetic field) and requires a large magnetic field gradient to affect the solutal buoyancy driven flow. We developed a novel flow control methodology wherein we make the protein solution paramagnetic (attracted to a magnetic field) by adding paramagnetic salts. Fairly high paramagnetism can be induced by adding small quantities of the salt. We successfully grew protein crystals from these solutions and developed the theory for magnetic flow control. Validation experiments are planned for early 2005. A major advantage of the technique is that we can now do the magnetic flow control experiments in-house at NASA Marshall Space Flight Center using the available 5 Tesla magnet. The research on the project has resulted in several presentations, papers, and awards. We also developed key collaborations in applying our research to artificial gravity applications for supporting NASA's extended duration exploration missions. The concept is to use magnetic levitation as a microgravity model for cell cultures in order to determine the gravity threshold levels and artificial gravity requirements for exploration purposes. Task Objectives Specific goals for the investigation are: 1. To develop an understanding of convection control in diamagnetic fluids with concentration gradients through careful experimentation. Specifically solutal buoyancy driven convection due to crystal growth will be considered. Strong magnetic fields and field gradients will be utilized to unambiguously resolve the role of fluid flow in the process. 2. To develop predictive measures for successful crystallization in a magnetic field using analyses and numerical modeling for use in future protein crystal growth experiments. This will establish criteria that can be used to estimate the efficacy of magnetic field flow damping on crystallization of candidate proteins. 3. To demonstrate the understanding of convection damping by high magnetic fields to a class of proteins that is of interest and whose structure is as yet not determined. 4. To compare quantitatively, the quality of the grown crystals with and without a magnetic field. X-ray diffraction techniques will be used for the comparative studies. Prior ambiguities of space-based protein crystallization need clarification. Understanding the fundamentals of a process is critical to major innovations and breakthroughs. Protein crystallization is however a complex process involving the interactions of several physical phenomena such as kinetics, fluid flow, physical chemistry, self-assembly, etc. and is still an empirical science. It holds the promise of the development of next generation designer drugs for and hence it is vigorously pursued by the drug industry. Protein crystallization experiments have flown on more space missions than any other targeted research effort but a comprehensive understanding of the fundamental processes and the beneficial effects of the microgravity environment are not clear. This investigation uses magnetic fields to counter gravitational effects – a terrestrial simulation of microgravity, in order to investigate crystal growth phenomena in reduced gravity. The current working hypothesis is that convection is detrimental to macromolecular crystal growth, although the exact mechanisms are unclear, and yet only 20% of space-grown proteins have shown results better than ground-based studies. This investigation will clarify the impact of fluid flow during the growth process either as a causative factor of some undesirable effects or point to the need to investigate other (non-flow) effects to improve crystal quality. The study adopts a stepping stone approach to the problem by validating theory by simple experiments before moving on to full-fledged protein crystallization. It represents an important contribution in the understanding of the complex crystallization phenomena.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While a limited number of high quality crystals from space flights have been obtained, as the recent National Research Council (NRC) review of the NASA microgravity crystallization program pointed out, the scientific approach and research in crystallization of proteins has been mainly empirical yielding inconclusive results. This research will answer some fundamental questions with regard to fluid flow and its effects in protein crystallization. The understanding of fluid flow control using magnetic fields and gradients is a research topic of importance as well with potential reduced-gravity applications. As a specific application to protein crystallization studies, if the ground-based experiments show crystal quality improvement, then the project could lead to space-based experiments that require much weaker magnetic fields.
Task Progress:	Several significant accomplishments were recorded in the project in 2004. These are itemized below: 1. A second series of flow control experiments in paramagnetic fluids were completed using a 5 Tesla magnet available at NASA Marshall Space Flight Center (MSFC). Custom hardware for optical Schlieren/shadowgraph diagnostics was built and utilized for this purpose. The theory developed for complete flow reversal of a paramagnetic salt solution was verified in the experiments. Detailed instability analysis of the flow was completed and corroborated by experiments. 2. One paper on theory and experiments has been published in the Journal of Crystal Growth and another paper on instability analysis was written and submitted for publication. 3. Two papers (one an invited keynote speech) on the studies were presented at two International Conferences. 4. Susceptibility and density measurements as a function of protein concentration were completed at MSFC and Brown University. Preliminary Lysozyme protein crystallization experiments were completed at Brown University. 5. Based on the above data, limits of convection control using magnetic fields during conventional (diamagnetic solution) Lysozyme protein crystallization were established. It was determined that a g-level of only ~0.2 go could be simulated. This g-level is considerably larger than 10-6 go (microgravity) that represents the steady state environment on the International Space Station. 6. A novel technique of magnetic convection control in the above system using a paramagnetic salt (MnCl2.4H2O) was formulated, crystals grown, and flow control theory developed. The importance of this finding is that the crystal growth

	solution now becomes paramagnetic allowing us to test flow control in protein crystallization using the in-house (MSFC) 5 T superconducting magnet. Experimental verification will be done in early 2005. 7. Thaumatin protein purification and crystallization protocols were completed and crystals were successfully grown. This will also be subsequently used as a benchmark protein to test flow control in protein crystallization. 8. Second generation flow and diagnostics apparatus to study protein crystal growth convection control in magnetic fields was built. 9. Several awards and honors were received (listed in a later section) and key collaboration was developed in applying the knowledge from the project to artificial gravity simulation for supporting NASA exploration goals.
Bibliography Type:	Description: (Last Updated: 03/17/2017)
Articles in Peer-reviewed Journals	Ramachandran, N., and Leslie, F.W. "Using Magnetic Fields to Control Convection during Protein Crystallization - Analysis and Validation Studies" Journal of Crystal Growth, Vol 274/1-2 pp 297-306. , Jan-2005
Awards	Ramachandran, N., and Leslie, F.W. "NASA MSFC Director’s Commendation for dedicated service and technical contributions and advances in Research " Jul-2004
NASA Technical Documents	Ramachandran, N., and Leslie, F.W. "A technique for rapidly producing a fluid concentration gradient in a microgravity environment using magnetic fields" NASA Space Act Award and Certificate. General audience Aug-2004
Presentation	Ramachandran, N., and Leslie, F.W., "Magnetic Control of Convection during Protein Crystallization" 14th International Conference on Crystal Growth, Grenoble, France. Aug-2004
Presentation	Ramachandran, N. "Using Strong Magnetic Fields to Control Solutal Convection" Microgravity Transport Processes in Fluid, Thermal, Biological and Materials Sciences Conference III, Davos, Switzerland (Keynote Talk) Sep-2003
Presentation	Ramachandran, N. "Controlling Solutal Convection using Strong Magnetic Fields" Science Directorate Seminar, NASA MSFC. Jun-2004
Presentation	Ramachandran, N. "Using Magnetic Fields for Simulating Reduced Gravity Environments" Science Directorate Presentation. Sep-2004